

AYUSH KOKANDE

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EDUCATION

Courant Institute of Mathematical Sciences

New York, United States

M.S. Computer Science (Concentration: Artificial Intelligence) — Current CGPA: 3.50/4.0

Aug 2025 – Aug 2027

Netaji Subhas University of Technology

New Delhi, India

B.Tech., Computer Engineering — CGPA: 3.72/4.0

Aug 2019 – Jul 2023

PROJECTS

Reasoning Model Post-Training: SFT + GRPO | PyTorch, vLLM, SLURM *Mar 2026 – Apr 2026*

- Built an end-to-end SFT pipeline for **Qwen2.5-Math-1.5B** on PrimeIntellect **INTELLECT-MATH** reasoning traces, i.e. **response-masked NLL** loss, BF16 **AdamW** with linear warmup, gradient clipping at **1.0**, and gradient accumulation across a **subset-size sweep** {128, 256, 512, 1024, *full*}; ran on a 2-GPU **trainer↔vLLM topology** that hot-reloads policy weights for periodic **MATH-12K** held-out eval, with decoupled **train_step/eval_step** curves in **wandb**.
- Implemented a from-scratch **GRPO** loop on **Countdown** (10K train / 1024 val) with **vLLM group rollouts** ($G = 8$, $T = 0.7$), **group-normalized advantages**, microbatched policy-gradient updates, and **PPO-style importance-ratio clipping** at $\epsilon = 0.2$ for the off-policy variant; tuned the learning rate via a **3-config sweep** and pushed validation reward past the **30% Countdown target**.
- Ran end-to-end **GRPO ablations** comparing no-baseline REINFORCE vs. **REINFORCE + group-normalized baseline**, **masked_mean** vs. **Dr. GRPO-style masked_normalize** sequence-level credit assignment, and group-std normalization on vs. off; compared validation reward and gradient-norm stability across configs.
- Decomposed reward into **format vs. answer** components and logged **per-token entropy**, **clip-fraction**, **grad-norm**, and reward variance to **wandb**; packaged the pipeline as **reproducible SLURM sbatch jobs** with a uv-managed Python environment, deterministic seeds, and versioned checkpoints, and **unit-tested** tokenization, response masking, group-normalized rewards, and the PPO-clip objective with **pytest**.

Long-Context Jailbreak Interpretability | PyTorch, Transformers, Accelerate *Mar 2026 – May 2026*

- Extracted a causal **refusal direction** from **Qwen3-14B** (40 layers, $d_{model} = 5120$, 32K context) via **difference-of-means** over **256 harmful AdvBench + 256 harmless Alpaca** activations, localizing the safety read-off at **layer 20** and causally validating via layer-wise directional ablation that drives the harmful-refusal rate from **95% → 1%**.
- Designed and ran a **preregistered two-arm context-length scaling sweep** at $L \in \{0, 512, 2K, 8K, 16K, 32K\}$ benign-filler tokens (intact vs. directionally-ablated $\hat{V}_{refusal}$), **falsifying the dilution hypothesis at this scale** — intact-arm refusal stayed **flat at ~95%** across all L while the ablated control collapsed to **~1%**, validating methodology and turning a negative finding into a clean falsifiable result.
- Built a **validity battery** addressing a domain-shift critique on $\hat{V}_{refusal}$: a **covariate-matched** harmful/harmless re-derivation (verb-class and word-length matched) with per-layer cosine comparison to the original, and a preregistered **2×2 intent × topic ANOVA** (50 prompts/cell, decision rule on partial η^2 at layer 20) ruling out vocabulary and topic-dominance confounds in the learned refusal direction.
- Packaged the full pipeline as **reproducible SLURM batch jobs** on **2×A100 (SXM4-40GB)** inside a **Singularity** container with a uv-managed Python environment and **bf16** inference; per-prompt projections, cell-level ANOVA outputs, and figures emitted to versioned results dirs for paper-ready artifacts.

End-to-End Transformer LM Architecture | Python, PyTorch, NumPy

Jan 2026 – Feb 2026

- Built a **17M-parameter decoder-only Transformer** in PyTorch **from scratch** — RMSNorm, RoPE, SwiGLU, and pre-norm blocks **without using nn.Transformer or nn.MultiheadAttention**, implementing every layer from tokenizer to **lm_head**.
- Trained a byte-level **BPE tokenizer (10K vocabulary)** on the full TinyStories corpus in **under 2 minutes** via multiprocessing pre-tokenization and an incremental pair-count index that restricts merge updates to affected words.
- Implemented **AdamW** with decoupled weight decay, **cosine LR annealing with linear warmup**, global-norm gradient clipping, memory-mapped data loading, and resumable model + optimizer checkpointing.

- Ran matched-parameter ablations (**RMSNorm**, pre-norm vs. post-norm, **RoPE** vs. **NoPE**, **SwiGLU** vs. **SiLU**) to isolate each component's validation-loss contribution; validated all modules against a **reference suite** with tiktoken parity.

FlashAttention-2 & Systems Optimization | *Triton, nsys, PyTorch*

Feb 2026 – Mar 2026

- Authored a **fused Triton FlashAttention-2 kernel** with tiled Q/K/V blocking and online softmax, achieving an **18× forward speedup** vs. naive PyTorch on **A100 (SXM4, 40GB)**.
- Cut attention memory from $O(N^2) \rightarrow O(N)$ by streaming tiles through SRAM and recomputing statistics in the backward pass, enabling **65K+ sequence lengths** without OOM.
- Accelerated training of a **2.7B-parameter model** by **2×** through **BF16 autocast**, **torch.compile** integration, and kernel autotuning over block-size / num-warps configs.
- Profiled with **Nsight Systems** to isolate SM-occupancy stalls and warp-scheduling inefficiencies; remediated to reach **>90% of theoretical peak TFLOPS**.

EXPERIENCE

FinBox

Bengaluru, India

Software Development Engineer | *Go, Temporal, PostgreSQL*

Mar 2025 – Jul 2025

- Integrated workflow editor using serverless, enabling developers in teams across multiple domains to use the underlying framework to 10x their productivity
- Orchestrated Axis Bank's digital loan journey using **Temporal workflows**, enabling **200K+ monthly applications** while implementing distinct retry policies that reduced manual intervention by **40%**.
- Optimized high-throughput backend services for clients including Muthoot Finance and Tata Digital, enhancing API performance by refining **Go concurrency patterns** and database locking strategies.
- Mentored and onboarded **3+ developers** through service documentation, pair programming, and structured walkthroughs.

Navi Technologies Pvt. Ltd

Bengaluru, India

Software Development Engineer | *Java, Kotlin, Kafka, ScyllaDB, PostgreSQL*

Jul 2023 – Sep 2024

- Architected a high-availability Customer Device Service using **Kotlin**, **Spring WebFlux**, and **Reactor Kafka**, achieving a non-blocking I/O pipeline that supported **10x traffic growth** to **44k+ peak IOPS**.
- Engineered a migration strategy for **600GB+** of data from PostgreSQL to **ScyllaDB**, utilizing **S3 bulk loading** to bypass application-layer bottlenecks and reduce data migration costs by **60%**.
- Redesigned legacy relational schemas into optimized **NoSQL single-table designs**, implementing a **dual-write architecture** to ensure strong consistency for user mappings while maintaining eventual consistency for analytical queries.
- Implemented fraud-AML retry workflows and exposed PAN Match data using Levenshtein thresholds on the VKYC Portal, reducing AHT from **3–4s to 2s**.
- Enabled CKYC upload exclusively for OKYC-approved customers, reducing escalations to **zero** and lowering OCR processing cost by deprecating the HyperVerge API.

Samsung Research & Development Institute

Bengaluru, India

Software Development Engineer Intern | *C++, Google Test*

May 2022 – Jul 2022

- Improved unit test coverage for ISM modules from **57% to 80%** using GoogleTest and targeted testing strategies.
- Collaborated with senior engineers to ensure high-quality and timely delivery of module enhancements.

TECHNICAL SKILLS

Languages: Python, Java, Kotlin, Go, C/C++, SQL, NoSQL

ML / Post-Training: PyTorch, Transformers (HuggingFace), vLLM, Triton, Accelerate, wandb, SLURM, CUDA

Backend Frameworks: Spring Boot, Spring WebFlux, Reactor Kafka, Temporal, Netty, Node.js

Developer Tools: Git, Docker, IntelliJ, AWS S3, uv, Singularity

Databases: PostgreSQL, ScyllaDB, Cassandra, DynamoDB

Concepts: SFT, GRPO / PPO, RLHF, Mechanistic Interpretability, Distributed Systems, Microservices, CI/CD